```python
graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 2), ('E', 4)],
    'C': [('F', 2)],
    'D': [],
    'E': [('F', 1)],
    'F': []
}
h = {'A': 5, 'B': 4, 'C': 3, 'D': 2, 'E': 1, 'F': 0}
def astar(start, goal):  1 usage
    open_list = [(h[start], 0, start, [start])]
    while open_list:
        open_list.sort()
        f, g, node, path = open_list.pop(0)
        if node == goal:
            print("Path:", path)
            print("Cost:", g)
            return
        for neighbor, cost in graph[node]:
            new_g = g + cost
            new_f = new_g + h[neighbor]
            open_list.append((new_f, new_g, neighbor, path + [neighbor]))
astar( start: 'A',  goal: 'F')
```

```
Path: ['A', 'C', 'F']
Cost: 5

Process finished with exit code 0
```